FEATURE

Live-imaging sonar use in Texas crappie fisheries: Examining population-level responses due to potential increases in exploitation

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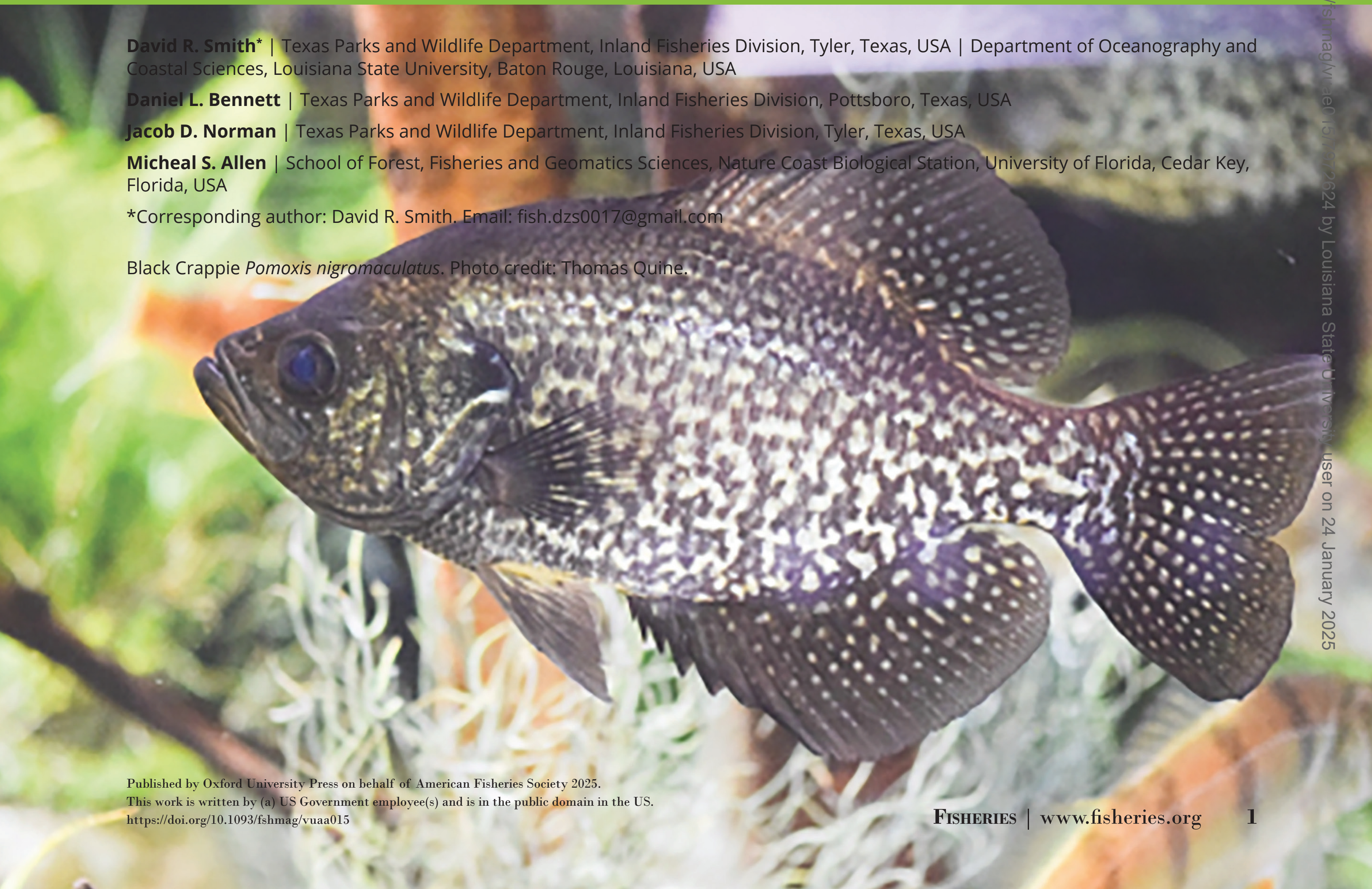

Black Crappie *Pomoxis nigromaculatus*. Photo credit: Thomas Quine.

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ABSTRACT

The use of live-imaging sonar (LIS) has surged in popularity among anglers in recent years, especially those targeting crappie *Pomoxis* spp. With LIS, anglers have perceived an increase in catchability of crappie and concerns have been raised about the resilience of crappie populations. As an attempt to provide information in response to these concerns, our study addressed the potential for LIS to increase angler catch rates and exploitation rates, with implications for potential overfishing. Using creel survey data collected on three regionally significant crappie fisheries, we subset creel estimations by LIS use (i.e., users and nonusers) for anglers targeting crappie. Users caught and harvested crappie at significantly higher rates than nonusers. Results suggest that LIS use has the potential to increase overall exploitation rates among the study reservoirs. Although exploitation may increase, we found that the potential for overfishing was minimal, given fishing mortality estimates as high as 79% and a 254-mm minimum length limit. Our study provides a baseline evaluation of current LIS use within popular crappie fisheries while showcasing how increasing fishing mortality could affect the sustainability of crappie populations across a range of growth and exploitation scenarios.

INTRODUCTION

A primary component of adaptive fisheries management requires continually refining management strategies while adjusting to changes in fish populations and angler behavior (Pope et al., 2014). In the current fisheries landscape, information is becoming more accessible (Bradley et al., 2019; Cooke et al., 2021), the use of emerging technologies is increasing (Cooke et al., 2021; Feiner et al., 2020), and angling techniques continue to advance (Lennox et al., 2017; Meals et al., 2012). Such shifts in angler behaviors can lead to improvements in catch and harvest efficiencies that affect fish populations and, in some cases, have led to overexploitation (e.g., Parkinson et al., 2004; Post et al., 2002). Innovations in modern angling sonar present a primary example of a shifting technology, as recent advancements have improved angler abilities to locate fish, observe movements, and detect strikes (Cooke et al., 2021; Feiner et al., 2020; Neely et al., 2023b).

Recreational angling sonar has improved considerably since these devices were first introduced in the late 1950s, with major advancements in angling sonar capabilities occurring over the most recent decade. Modern angler sonar devices now employ advanced transducers with multiple beams emitting different frequencies to produce extremely accurate two-dimensional images of fish and underwater environments (Garmin, 2023; Humminbird 2023; Lowrance 2023; Neely et al., 2023b). The most recent innovation in angling sonar has led to the introduction of live-imaging sonar (LIS), which allows anglers to locate fish and observe individual fish behaviors in relation to the presence of a bait. In 2018, LIS was effectively introduced to anglers with Garmin LiveScope (Garmin, 2023), and several manufacturers in the following years have introduced similar models employing LIS (e.g., Humminbird MEGA Live Imaging; Lowrance ActiveTarget).

Since 2018, this technology has been adopted by anglers throughout North America, particularly among those targeting crappie species (defined as White Crappie *Pomoxis annularis* and Black Crappie *P. nigromaculatus*). As crappie catch and harvest efficiencies were presumptively improved with the introduction of LIS, angler concerns have been voiced by several media sources across the Southeast and Midwest of the USA (e.g., Anthony, 2022; Cloud 2022; Hrenchir, 2022). Crappies exhibit a high level of site fidelity (Fryda et al., 2008; Stewart et al., 2014) and seasonally aggregate (Isermann et al., 2010; Phelps et al., 2009; Reed & Pereira, 2009) allowing anglers to locate schools of fish using sonar (e.g., Brown, 2021). Further, crappie fisheries are among the most intensively harvested freshwater fish species in the USA (Miranda, 1999) and often require regulatory action to prevent overfishing (Dotson et al., 2009) and maximize yield (Allen et al., 2013).

To date, the effects of LIS on angler catch and harvest rates for crappie fisheries have seldom been examined in the published literature; although, interest in the topic has recently increased. While not currently published, reports of an angler survey conducted by the Arkansas Game and Fish Commission demonstrated that anglers using LIS were harvesting crappies at lower rates than LIS nonusers, although overall catch rate was improved (Zellers, 2022). A study in Kansas also found that differences in crappie angler catch rates between LIS users and nonusers were nonsignificant (Neely et al., 2023b). However, the extent of this study's findings remain limited as it characterized an opportunistic sample of anglers across a single reservoir during 2 d of angling.

Given the scope of prior studies, there is a need to evaluate the impacts of LIS across a wider breadth of angling avidities, broader spatial scales, and different fishing conditions. The objectives of this study were to incorporate an LIS evaluation at popular crappie fisheries by (1) measuring the extent of LIS use among anglers directly targeting crappie, (2) quantifying the differences in catch and harvest efficiencies between LIS users and nonusers, and (3) estimating potential population-level impacts of inflated exploitation to examine the overfishing potential of LIS.

METHODS

Creel data

Annual creel surveys were conducted on three popular Texas crappie fisheries between June 1, 2022, and June 1, 2023, to obtain estimates of catch, harvest, and angler effort (hours fished) for anglers directly targeting crappies. Study sites included Lake Fork, Lake O' the Pines, and Sam Rayburn Reservoir, all located in eastern Texas and comprising some of the largest and most popular crappie fisheries in the state.

Each creel survey was conducted according to Texas Parks and Wildlife Department (TPWD) protocol based upon Malvestuto et al. (1978) and Pollock et al. (1994, 1997). All creel surveys were conducted during daylight hours and the spatial sampling frame was defined as the entire reservoir. Sampling was random, stratified by day type (weekday or weekend) and quarter (i.e., three consecutive months). Surveys were conducted on a minimum of five weekend days and four weekdays per quarter. Holidays falling on weekdays were treated as weekend day. The selection of sampling days to generate quarterly estimates were made following procedures outlined by Pollock et al. (1994).

Roving creel surveys were conducted on Sam Rayburn, while access point creel surveys were conducted at Lake O' the Pines and Lake Fork. Harvest and catch rates were calculated for roving surveys using the mean of individual ratios estimator and for access point surveys using the ratio of means estimator (Hoenig et al., 1997; Pollock et al., 1997) within a standardized creel analysis program written in SAS by one of the pioneers of creel survey research, S. P. Malvestuto. It is well established that differential estimation procedures between roving and access point surveys can lead to biases when comparing creel metrics across methods (e.g., Alexiades et al., 2015; D. R. Smith et al., 2024). In this study, differences were computed across methodologies based strictly on the proportional difference in harvest between LIS users and nonusers, such that comparisons were valid across survey methods.

During creel surveys, we collected traditional creel data (e.g., fish harvested, hours fished) as well as whether LIS was used during the angling trip (hereafter referred as "users") or not used during the trip (hereafter referred as "nonusers"). Our study was focused on evaluating how LIS has affected directed crappie anglers; thus, creel data applied in this study did not include anglers targeting other species or all species (e.g., anything that bites). Catch and harvest rates were compared between users and nonusers using a Dunn multiple comparison test (Midway et al., 2020) and the proportional size structure of harvested fish was compared using a Kolmogorov–Smirnov test ($\alpha = 0.05$ for both tests).

Aggregate estimates of harvest rate and catch rate were calculated across all three reservoirs using a generalized additive model framework (Wood, 2006) as the relationship between the number of fish caught or harvested with effort (hours fished) was nonlinear (e.g., Robertson et al., 2024), reaching an asymptote at the upper end of effort values. Models were built with the response as the number of fish caught or harvested by an angling party and the predictor as the number of hours fished per angling party. Response variables were count data that exhibited overdispersion; therefore, models were run using the negative binomial distribution. A random effect was applied for the LIS term and nested within each reservoir, allowing the amount of smoothing to vary by reservoir. To account for variability by day type (i.e., weekend or weekday), a random effect was applied in the aggregate model for this term (e.g., Pedersen et al., 2019). Predictions were derived from aggregate models to estimate mean catch and harvest rates per angling hour and associated standard errors. All statistical analyses were performed using R (R Core Team, 2020) and generalized additive models were built using the package *mgcv* (Wood, 2006).

Differential harvest between *users* and *nonusers* was calculated using the following equations:

$$H_i = Scale \times Hrate_i, \tag{1}$$

where H_i is the estimated harvest for the LIS term, i (i.e., user or nonuser); $Scale$ is a scaling term set equivalently between the LIS term such that direct comparisons were estimated (i.e., an equivalent amount of effort for both user groups); and $Hrate_i$ is the estimated aggregate harvest rate (harvest/hour). Proportional harvest (P_i) was calculated for each group as:

$$P_i = \frac{H_i}{H_{user} + H_{nonuser}} \tag{2}$$

and used within subsequent estimations to determine potential impacts on fishing mortality.

Exploitation

To simulate how exploitation rates may rise due to LIS, we estimated inflated exploitation as:

$$U_{inflate} = \left(1 - \left(2P_{nonuser} - 1\right)\right) * U_{prior}, \tag{3}$$

where $U_{inflate}$ is the inflated exploitation rate, $P_{nonuser'}$ is the proportional harvest of nonusers, and U_{prior} represents a theoretical exploitation rate prior to the introduction of LIS. It is important to distinguish here that exploitation rate (U; i.e., the proportion of fish within a population that experience mortality due to fishing) is not synonymous with harvest rate (i.e., the number of fish harvested per angler hour). In this equation, 1 represents current harvest trends (i.e., the sum of $P_{nonuser}$ and P_{user}) and $2P_{nonuser}$ represents theoretical harvest given users did not exist in the fishery.

While an exact estimation of fishing mortality within Texas crappie populations prior to the introduction of LIS was preferred, these data were not available. To calculate inflated exploitation rates, we used three surrogate measures of nonuser exploitation ranging from 30% (low) to 50% (moderate) to 70% (high). The mean fishing mortality estimate derived from 18 different crappie exploitation studies in Allen et al. (1998) was 48%, such that our estimates of exploitation are measures that reflect the strong harvest-orientation of typical crappie fisheries (Miranda, 1999). Although exploitation levels above 70% are uncommon (only one such study in Allen et al., 1998), we included this scenario to quantify potential impacts for crappie fisheries where high exploitation levels occurred prior to the onset of LIS. Hereafter, the initial surrogate values of exploitation used in this study will be referred to as nonuser exploitation.

Age-structured model

We built an age-structured equilibrium population model (ages 1–8) in R (R Core Team, 2020) similar to that described by Dotson et al. (2009), but parameterized for White Crappie using fast, moderate, and slow growth rates (Table 1). While harvest estimates were derived combining both White Crappie and Black Crappie, these species are traditionally managed as a single stock and hybridize in many natural populations (S. M. Smith et al., 1994; Travnichek et al., 1996). Combined estimates using both species are commonly applied when evaluating crappie fisheries (e.g., Allen et al., 1998; Isermann et al., 2002; Meals et al., 2012).

We parameterized this model for White Crappie as the species typically exhibits faster growth rates than Black Crappie, and it was important that our simulations did not underestimate average growth rates. Moderate growth rates were taken from Allen and Miranda (1995), where the average length at age was computed across several agency reports, theses, and journal articles. Slow growth rates were derived using length at age estimates 1 SD below moderate estimates (Allen & Miranda, 1995). Fast growth rates were derived using historic TPWD age–length data for White Crappie compiled from five Texas reservoirs (Waco, Lavon, Harris, Limestone, Aquilla).

While data from our study reservoirs was preferred, the quantity of age–length data collected for crappie on the study reservoirs was insufficient to build a growth models, such that we calculated growth rates using historic TPWD age–length data from the reservoirs listed above. Crappie growth rates in Texas are some of the fastest growth rates observed throughout their range, with fish reaching legal length (i.e., 254 mm) generally within age 1 or 2 (TPWD, unpublished data). Due to these growth rates exceeding 1 SD above moderate growth rates, we chose to use growth rates typically observed in Texas crappie fisheries, such that estimates could be transferrable to the study reservoirs.

Age-structured models were built nearly identically to Dotson et al. (2009), within the R programming language rather than Excel (in our GitHub repository, we have included an Excel spreadsheet following the Dotson et al., 2009 approach). We suggest readers interested in further details about our age-structured modeling approach to consult Dotson et al. (2009), our extended methods section (Supplement 1), and our GitHub repository (https://bit.ly/40DlMQ0).

All simulations were run across a 100-year timeline. In the first year, N_1 was set to 10,000 fish (R_0) and the number of fish in each subsequent age-class was calculated (Equation 1 in Supplement 1) to represent the unfished population, such that declines in each year-class were strictly from natural mortality. In years 2–100, N_a in each subsequent age-class declined due to fishing mortality and natural mortality (Equation 3 in Supplement 1). N_1 in years 2–100 was simulated following equation:

$$R_{eq} = R_0 \times Env, \tag{4}$$

where R_0 is the equilibrium number of age-1 recruits (10,000 fish) and Env was a term simulating environmental stochasticity with a mean of 1 following a log-normal distribution at a coefficient of variation of 0.8 to mimic the high recruitment variability typically exhibited by crappies (e.g., Allen & Miranda, 1998). To estimate an expected average response in relation to the effects of increased exploitation, mean spawning potential ratio (SPR), mean yield per recruit (YPR), and mean memorable percent were calculated as an average across years 50–100. Monte Carlo simulations were run repeating this timeline 10,000 times while calculating mean estimates under each level of exploitation for each growth rate.

RESULTS

Creel metrics

On two of the three reservoirs, crappie harvest rates ($P \leq 0.006$) and catch rates ($P \leq 0.026$) were significantly higher for LIS

Table 1. Definition and values of parameters used in a simulation model describing slow, moderate, and fast growth rates for crappie populations.

Parameter	Description	Value
	Growth (von Bertalanffy)	
L_∞	Asymptotic length in mm (growth: slow, moderate, fast)	333, 353, 356
K	Growth coefficient (growth: slow, moderate, fast)	0.325, 0.374, 0.691
t_0	Time (age, years) at zero length (growth: slow, moderate, fast)	0.174, 0.197, −0.056
	Natural mortality	
M	Instantaneous natural mortality rate (growth: slow, moderate, fast)	0.325, 0.374, 0.691
S_0	Finite annual natural survival rate (growth: slow, moderate, fast)	72%, 69%, 50%
	Fishing mortality	
$U_{inflate}$	Inflated recreational exploitation rate (exploitation: low, moderate, high)	34%, 57%, 79%
U_0	Total inflated capture rate (exploitation: low, moderate, high)	44%, 67%, 89%
D	Recreational discard mortality rate	9%
	Vulnerability	
TL_{cap}	Total length at 50% capture vulnerability (mm)	204
SD_{cap}	Standard deviation of 50% capture vulnerability (mm)	10
TL_{harv}	Total length at 50% harvest vulnerability (mm)	254
SD_{harv}	Standard deviation of 50% harvest vulnerability (mm)	13
TL_{mem}	Total length at 50% memorable size vulnerability (mm)	305
SD_{mem}	Standard deviation of 50% memorable size vulnerability (mm)	15
	Length-weight	
A	Length-weight coefficient	2.410×10^{-6}
B	Length-weight exponent	3.375
	Recruitment	
R_0	Average annual unfished recruitment	10,000
L_{mat}	Mean length at maturity (mm)	200
SD_{mat}	Standard deviation of length at maturity (mm)	20
W_{mat}	Mean weight at maturity (kg)	0.15

users compared to nonusers (Figure 1). However, on one of the study reservoirs (Sam Rayburn), differences between users and nonusers were nonsignificant for both catch rates ($P = 0.613$) and harvest rates ($P = 0.220$). The aggregate harvest rate estimated for users (1.65 fish/h; SE = 0.22) was significantly higher ($P = 0.027$) than the rate estimated for nonusers (1.26 fish/h; SE = 0.22). Catch rates followed a similar pattern with users (2.54 fish/h; SE = 0.19) catching fish at significantly higher rates ($P < 0.001$) than nonusers (2.05/h; SE = 0.19).

The proportional size structure of crappie harvest was significantly different for the LIS term ($P < 0.001$), with users harvesting significantly larger sized crappie on average (297 mm; SE = 0.69) compared to nonusers (287 mm; SE = 0.77). During creel surveys, fish lengths were collected by inch-class, and converted to mm for analysis such that the 10-mm difference we observed in average crappie harvested is important to view in this context. However, a majority of fish harvested by users were at memorable lengths (52%) with a higher proportion compared to nonusers (39%). Additionally, the proportion of fish harvested at trophy lengths (≥381 mm) by users (2.7%) was nearly sevenfold greater than the proportion observed by nonusers (0.4%).

Effort estimates for users either approached or exceeded a majority of overall directed crappie effort, ranging from 49% to 67% of total effort (Figure 2). In the aggregate, 57% of total directed crappie effort was observed by LIS users. The proportional harvest was disproportionately skewed toward LIS users, harvesting ≥ 69% of total crappies on Lake Fork and Lake O' the Pines. However, on Sam Rayburn Reservoir, users only harvested 44% of the total crappie harvest by anglers targeting crappie (Figure 2). In the aggregate, 61% of total crappie harvest was observed by LIS users.

Age-structured model

Implementing the observed harvest differences among user groups (Equation 3), fishing mortality increased by 14% relative to nonuser exploitation for low (increased from 30% to 34%), moderate (increased from 50% to 57%), and high (increased from 70% to 79%) levels of exploitation. The proportion of SPR simulations below the reference point of 30% SPR was negligible (<0.1%) under five of the six exploitation/growth scenarios. We observed the highest potential for overfishing under moderate growth rates and high exploitation levels; even so, the percentage of simulations under the overfishing

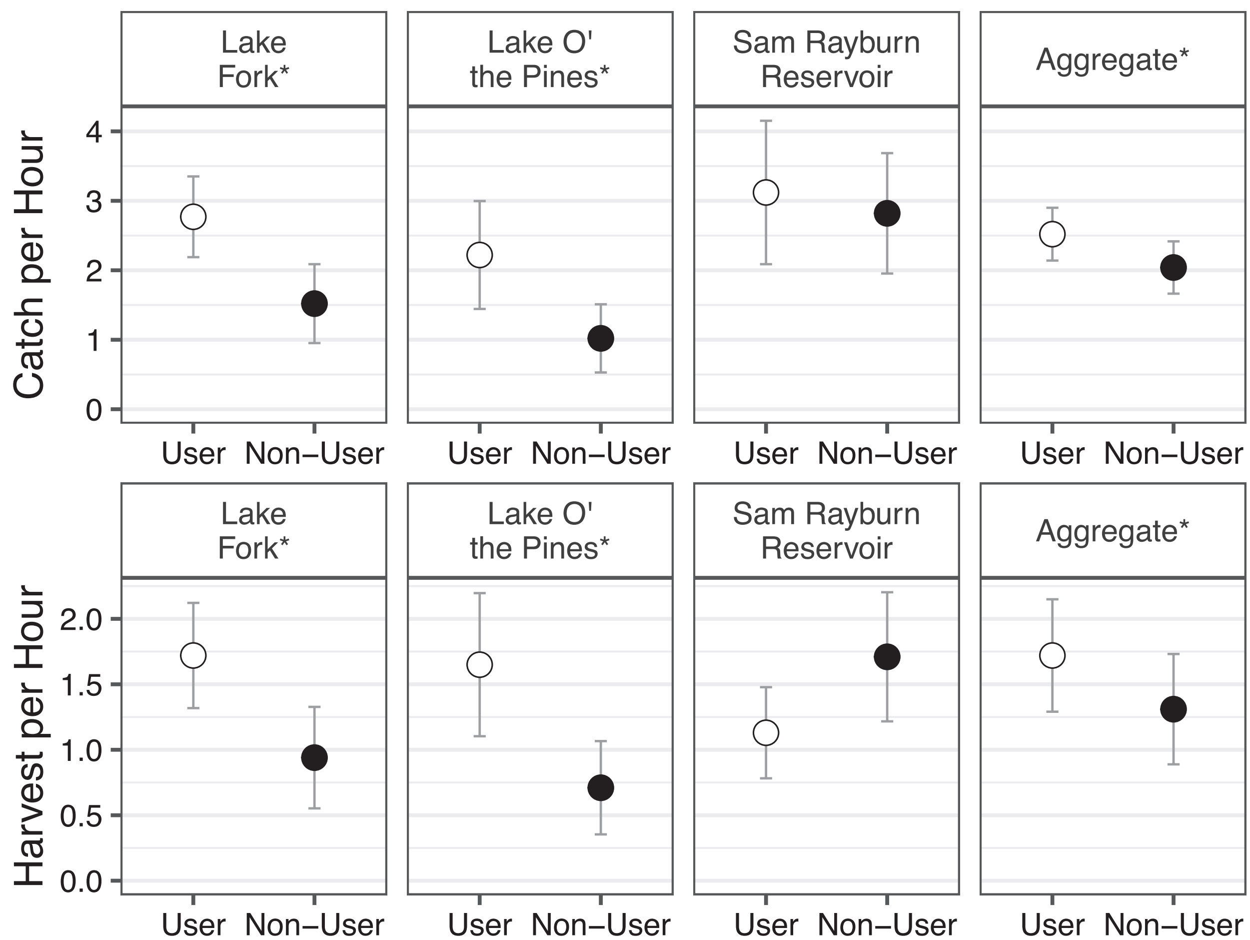

Figure 1. Estimated catch and harvest efficiencies for anglers that used live-imaging sonar during their angling trip (*user*) and anglers that did not use live-imaging sonar (*nonuser*). Error around each estimate represents 95% confidence intervals about the mean. Asterisks indicate significant catch rate or harvest rate differences between *users* and *nonusers* following a Dunn multiple comparison test. Aggregate estimates represent catch rate and harvest rate estimations for pooled data across all reservoirs within a generalized additive model.

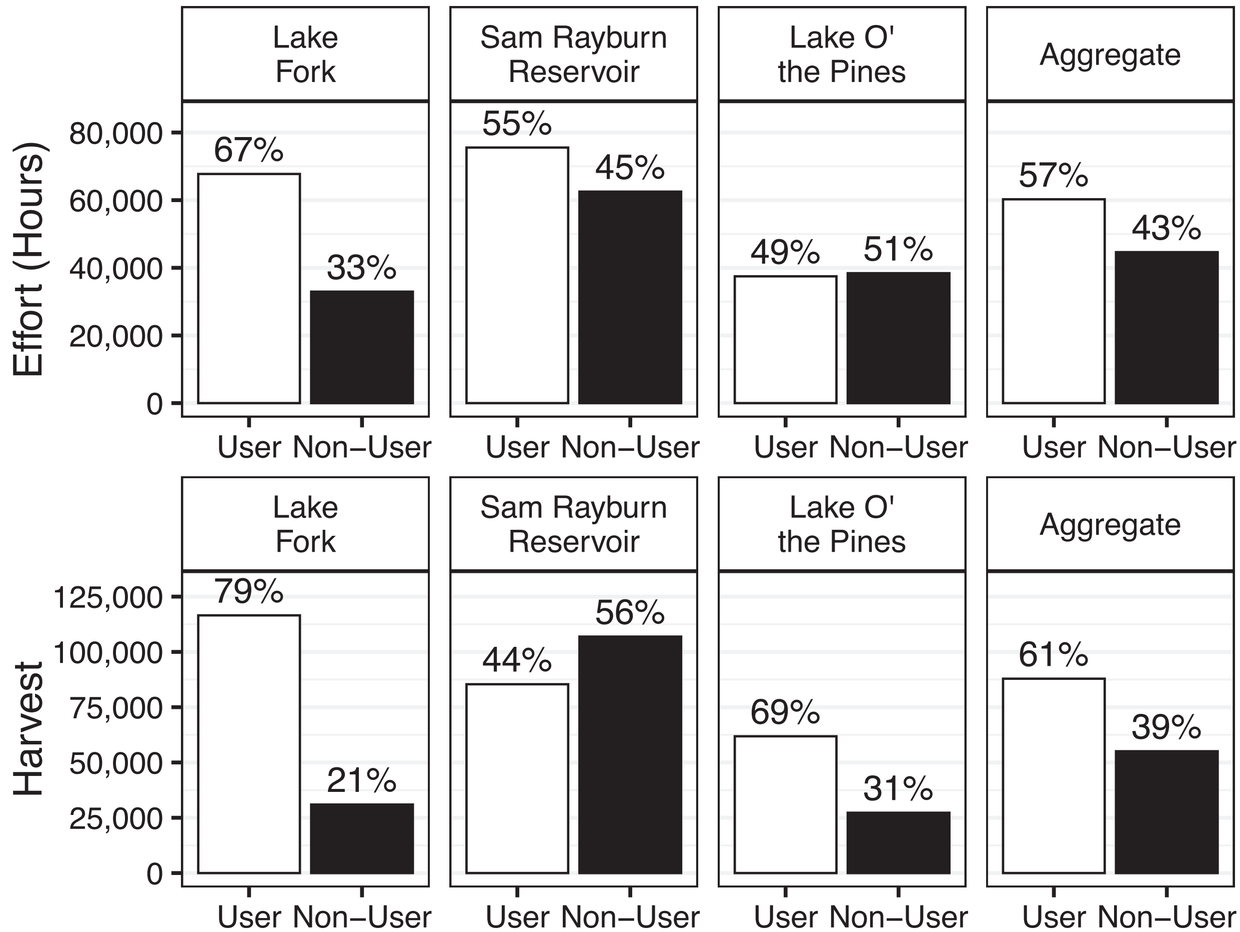

Figure 2. Estimated directed effort (angler hours) and crappie harvest for anglers directly targeting crappie. *Users* represent anglers that used live-imaging sonar during their angling trip and *nonusers* represent anglers that did not use live-imaging sonar. Aggregate estimates represent the mean estimate across all reservoirs.

threshold was less than 3%, illustrating the low likelihood of overfishing even at extremely high levels of fishing mortality (given the 254-mm statewide length limit in Texas).

Additionally, we identified the potential for growth overfishing as low under inflated levels of exploitation with median YPR estimates increasing with successively higher levels of exploitation. However, under moderate and slow growth scenarios experiencing high levels of exploitation, increases in median YPR from nonuser exploitation to inflated exploitation were minimal (<1%; Figure 4) illustrating that the maximum YPR for these growth scenarios may be near 79% and there may be potential for growth overfishing at fishing mortality levels only marginally higher than the levels used in this study.

Although yield continued to increase with higher levels of exploitation, our models estimated some population-level effects of increased fishing mortality as the proportion of memorable-sized crappie successively declined at higher levels of fishing mortality. At increasing levels of exploitation, we observed changes in the number of memorable-sized crappie in the population that may alter angling experiences. We observed a decline $\geq$ 6.7% in the median number of memorably sized crappie in the population under each growth scenario (Figure 5).

DISCUSSION

In a relatively short period of time since the effective introduction of LIS, results of this study suggest that LIS use has become prolific among Texas crappies. A likely reason for the high levels of LIS use we observed is that angler catch efficiencies have improved with LIS. Our findings illustrate that users caught and harvested crappie at significantly higher rates than nonusers. Feiner et al. (2020) found a similar increase in crappie harvest efficiency relative to the use of angling sonar, which suggests that the behavior and life history of crappies (e.g., Fryda et al., 2008; Phelps et al., 2009) may make this species particularly vulnerable to future innovations of angling sonar. However, the adoption of LIS across the study reservoirs was likely affected by a multitude of social factors not captured by this study (e.g., income, employment status). Further evaluations on LIS adoption rates may shed light on factors influencing the high level of LIS use observed on the study reservoirs. While results suggest that LIS use has the potential to increase exploitation rates, we found that none of the considered scenarios suggested that recruitment or growth overfishing is likely to occur given the existing size limit.

It is important to place the context of our findings within the regulatory framework of a 254-mm minimum length limit. For

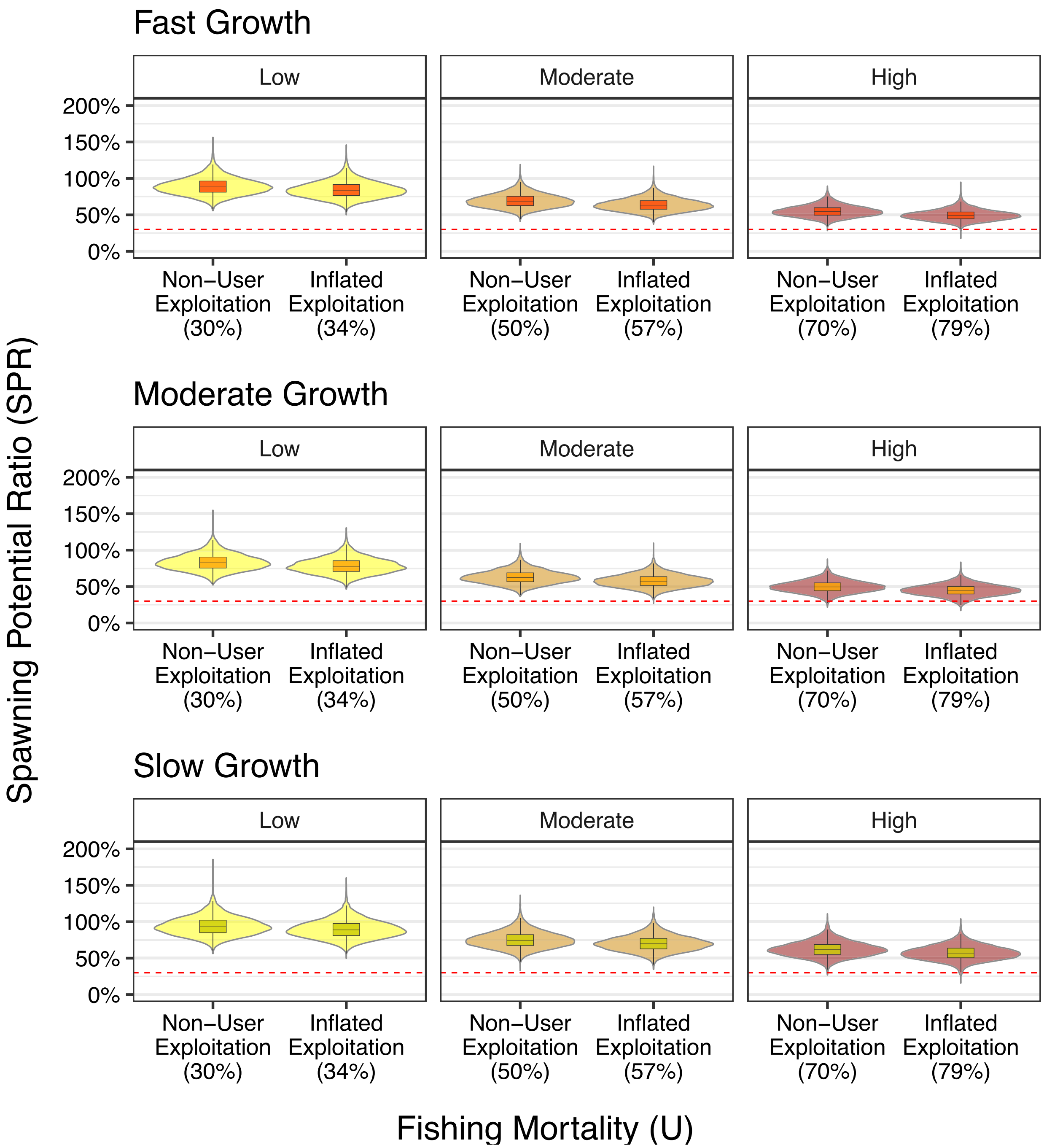

Figure 3. Simulations of spawning potential ratio (SPR) for slow, moderate, and fast-growing crappie populations based on low, moderate, and high levels of inflated exploitation presented as violin plots. Box plots indicate the quartiles centered around the median with whiskers indicating 1.5 times the interquartile range. Shading behind each box plot indicates the probability density with the width of the shaded area representing the proportion of simulations located across the range of data. The dashed horizontal line represents a SPR of 30% which was applied as a reference point indicative of recruitment overfishing in crappie populations. Nonuser exploitation estimates were set at three arbitrary starting points ranging from 20% to 35% to 50%. Inflated exploitation levels were then calculated using the differential harvest rates observed between *users* and *nonusers* for each level of nonuser exploitation.

fisheries with lower or no minimum length limit, the potential exists for overfishing (e.g., Dotson et al., 2009), and thus, LIS technology would be expected to increase the potential for overfishing. We ran additional simulations without a length limit and under high levels of inflated fishing mortality (79%). Within these simulations, the majority of moderate (94%) and slow growing crappie populations (62%) were below an SPR of 30%. Even without a minimum length limit, the potential for overfishing fast growing populations was low with <1% of simulations below our overfishing reference point.

While the potential for overfishing was minimal under all simulated levels of exploitation, our study illustrates that

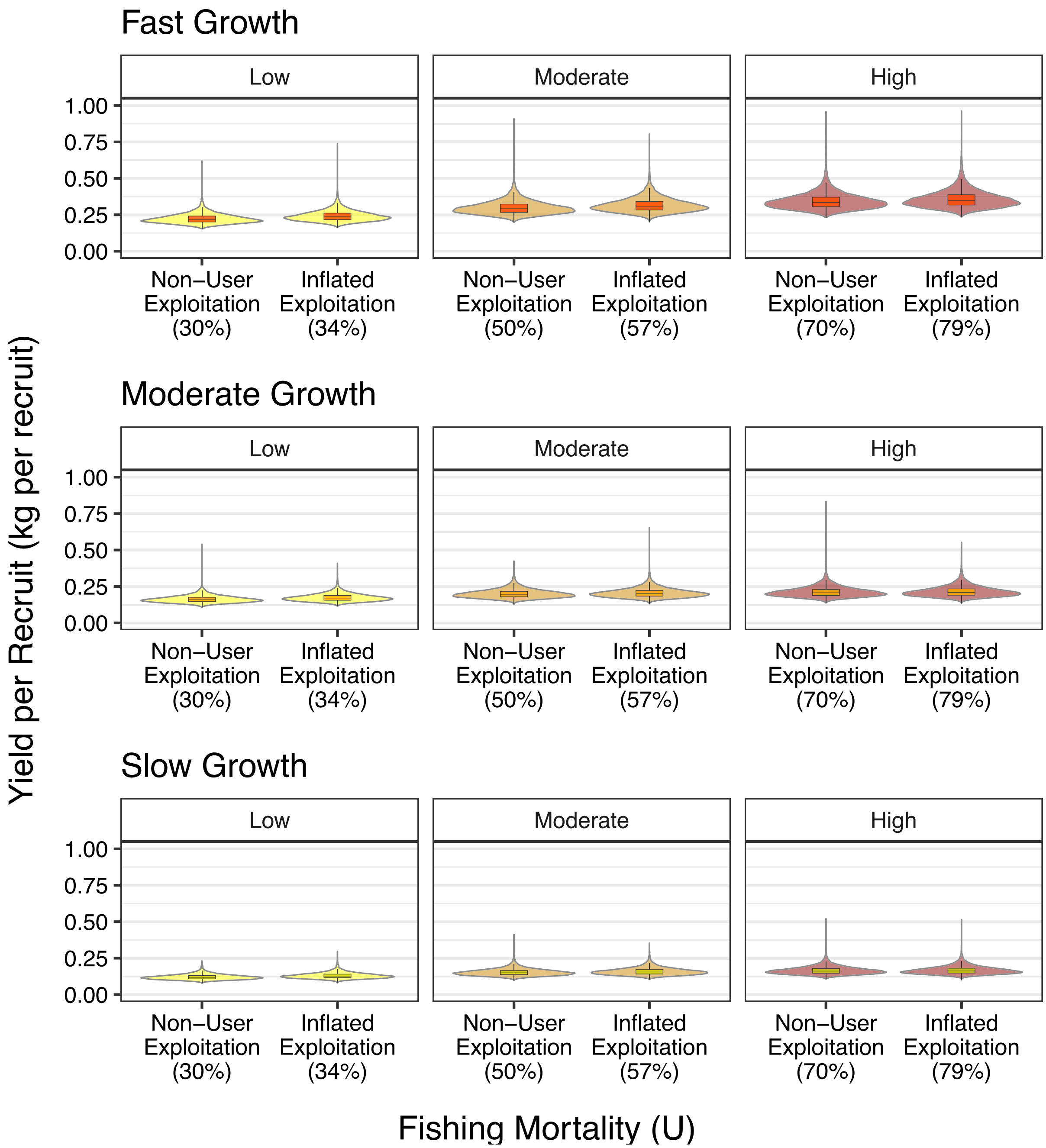

Figure 4. Simulations of the yield per recruit (biomass harvested per recruit) for slow, moderate, and fast-growing crappie populations based on low, moderate, and high levels of inflated exploitation presented as violin plots. See Figure 3 for a more detailed description of violin plot representation, nonuser exploitation, and full exploitation.

meaningful population-level impacts remain plausible as fishing mortality increases. Models estimated that abundance of memorable-sized fish declined with increasing levels of mortality; thus, making large fish rarer in the population, which can influence angler satisfaction. Given LIS users were harvesting significantly larger crappies than nonusers, the effects of LIS on the abundance of memorable sized fish were potentially underestimated as the size structure of capture vulnerabilities remained unchanged when comparing user groups. These factors are important considering that in many instances, the maximization of yield is not aligned with peak angler satisfaction (Cox et al., 2003) and the size of fish caught plays a meaningful role in determining angler satisfaction (Birdsong et al., 2021).

While our study indicated significant harvest disparities between LIS users and nonusers in the aggregate, a large amount of variability was exhibited in harvest trends by reservoir. For example, differences in harvest rates were nonsignificant relative to LIS use on Sam Rayburn Reservoir, which led to a minority of harvest observed by users (44%). In contrast, we observed substantially different harvest rates on Lake Fork, leading to 69% of overall harvest observed by users. Moreover,

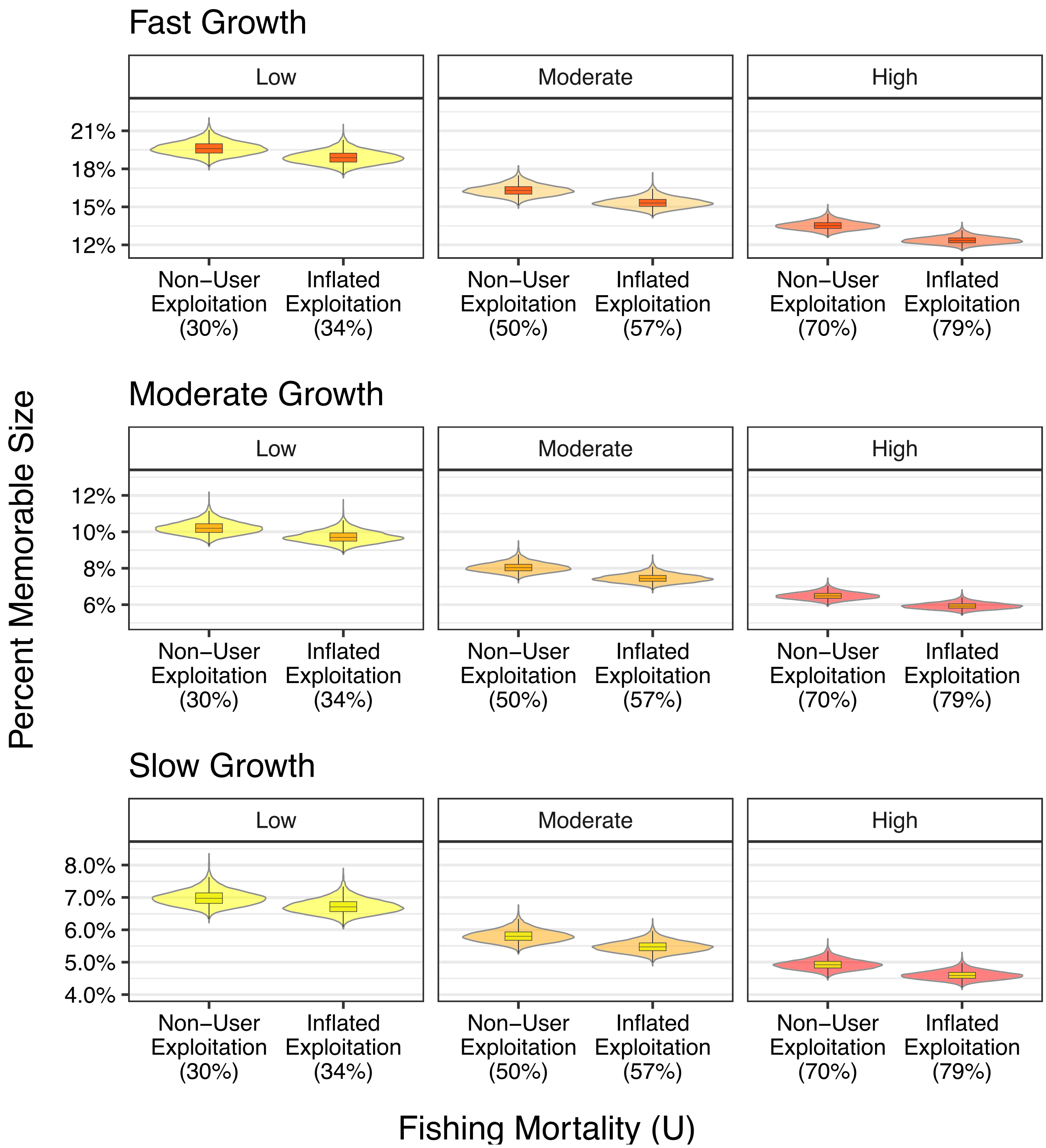

Figure 5. Simulations of the population at a memorable length (305 mm) for slow, moderate, and fast-growing crappie populations based on low, moderate, and high levels of inflated exploitation presented as violin plots. Simulations are presented as violin plots. See Figure 3 for a more detailed description of violin plot representation and inflated exploitation.

these two lakes also differ in respect to comparisons made with historic harvest rates suggesting that the impacts of LIS use on crappie populations may have distinct impacts across water bodies with varying characteristics. Further evaluations may be warranted to disentangle the relationship between LIS use and reservoir characteristics including biotic (e.g., aquatic plant coverage, trophic indices, forage abundance and diversity) and abiotic (e.g., bathymetry, standing timber coverage) factors relevant to the fishery. The results of our study were developed to provide a baseline characterization of LIS use in Texas and results suggest the relative impact of increasing fishing power due to LIS is likely to be system and context specific. Considering the impact of unique environmental conditions on each reservoir underscores the importance of developing reservoir-specific estimates of both exploitation and angler behaviors to inform management decisions.

Limitations

We did not gather any information on angler avidity, and it is possible that *users* represented a category of anglers with increased skill levels. However, aggregate directed crappie effort reached approximately 60% for users such that it is unlikely

LIS use was restricted to anglers with high avidities. It is possible that the most avid crappie anglers (e.g., chartered fishing guides, high-use anglers), who already caught and harvested significantly more crappie than the general angler, simply adopted the technology, and were identified as users in our study. Inferences were made based strictly on the harvest disparities we observed between users and nonusers, and further evaluations of avidity and LIS use may be warranted to truly separate the effects of skill and the use of this technology (e.g., Neely et al., 2023a).

Our simulations were performed assuming a constant rate of exploitation; however, catchability is influenced by fish population abundance (Post et al., 2002; Wilberg et al., 2009) such that constant levels of exploitation over time are unlikely with dynamic populations. Additionally, angler effort is correlated to angler catch rates (Cox et al., 2003) where shifting levels of catchability can lead to changes in a fishery that affect exploitation (e.g., Allen et al., 2013). We modeled each exploitation scenario based on a fixed growth rate; however, crappie populations typically exhibit density-dependent growth (Allen & Miranda, 1998; Pope et al., 2004) such that growth rates shift in accordance to changes in population size. Crappie populations also exhibit compensatory mortality such that increasing levels of exploitation can affect rates of natural mortality (Allen et al., 1998). While dynamic levels of exploitation, growth rate, and natural mortality were not accounted for in this study, developing a model to account for every source of stochasticity was not required for our primary research objectives. Although our models were a simplification of the complex environments exhibited within actual fisheries, results remain meaningful in the context of understanding an expected average response for crappie populations in relation to increased fishing mortality.

Management implications

The results of our study show that LIS use is prolific among Texas crappie anglers, and anglers that used LIS were on average harvesting larger-sized crappie with significantly improved catch and harvest efficiencies. While study results are specific to Texas reservoirs, these findings were from reservoirs where a high level of angling effort is directed toward crappies and similar patterns may occur on other popular crappie fisheries throughout North America. The results of this evaluation suggest that LIS users catch and harvest crappie at a greater rate than nonusers; however, it is not known if this has resulted in increased fishing mortality overall, or merely highlighted the differences in avidity and harvest practices of two distinct angler groups.

Although other studies have attempted to characterize the impact of angling sonar relative to angler catch and harvest efficiencies (e.g., Feiner et al., 2020), to our knowledge, our study represents a first attempt at predicting potential population level effects given increases in fishing mortality relative to increasing angling efficiency. Live-imaging sonar use only represents a single way that fishing power has changed over time. Future improvements are likely to occur in boat functionality (e.g., trolling motors, anchoring systems), tackle, angling techniques, and the accessibility of information, among other factors that can affect catch and harvest efficiencies. Emerging technology has the potential to continually improve angling efficiency, thus future studies evaluating the impacts of increasing fishing power relative to fish population dynamics will play an important role in exploring sustainable regulatory strategies.

SUPPLEMENTARY MATERIAL

Supplementary material is available at *Fisheries* online.

DATA AVAILABILITY

The data underlying this article will be shared on reasonable request to the corresponding author.

ETHICS STATEMENT

This study followed the ethical guidelines for publication provided by American Fisheries Society.

FUNDING

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CONFLICTS OF INTEREST

None declared.

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